

Ex.No:12

Date:13/10/25

TO CAPTURE, SAVE, AND ANALYZE NETWORK TRAFFIC USING WIRESHARK TOOL

Aim:

To capture, save, and analyze network traffic on TCP/UDP/IP/HTTP/ARP/DHCP/ICMP /DNS using Wireshark Tool.

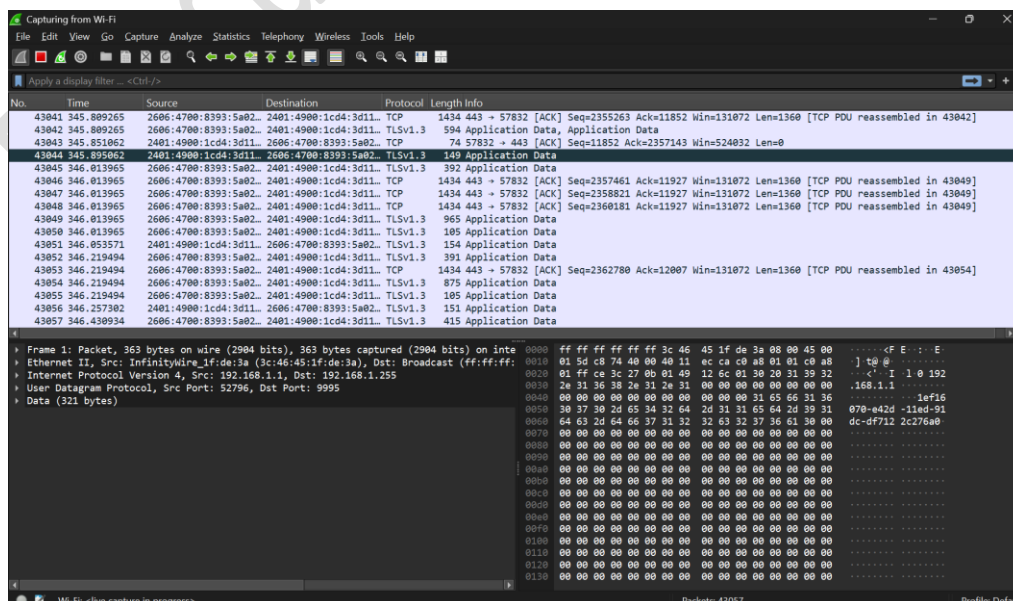
Procedure:

1. Install Wireshark

- Download and install Wireshark from <https://www.wireshark.org/>
- Make sure to install the required packet capture drivers (e.g., WinPcap or Npcap on Windows).

2. Start Capturing Traffic

- Open Wireshark.
- Select the correct network interface for capturing (Wi-Fi, Ethernet, etc.).
- Click on the interface to start live capture.

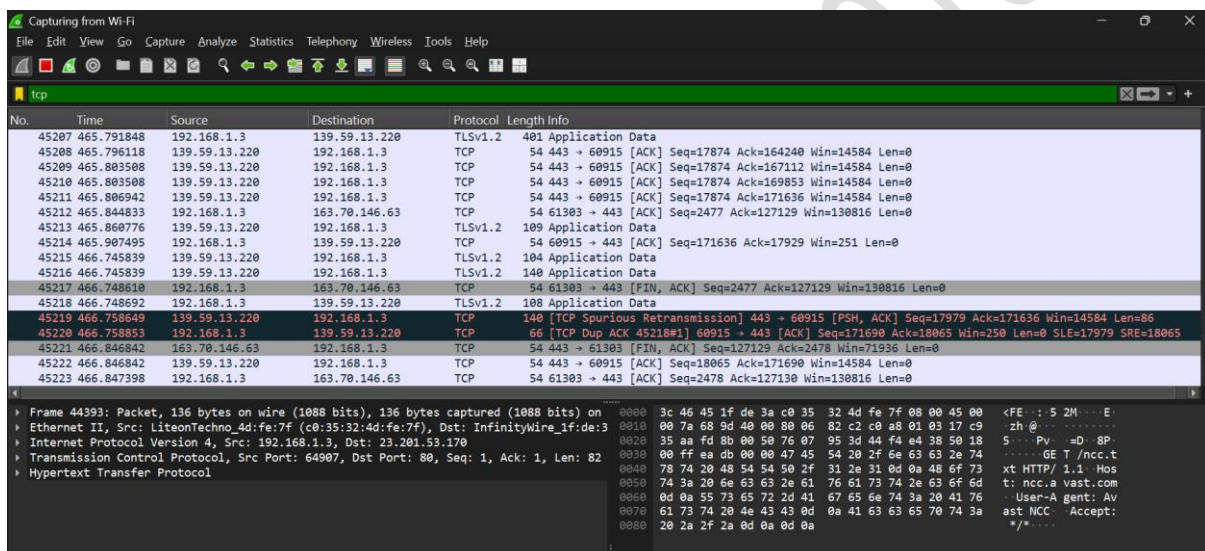


3. Apply Capture Filters

To reduce the amount of data captured, you can apply capture filters.

Example filters for protocols:

- TCP only: tcp
- UDP only: udp
- ARP only: arp
- ICMP only: icmp
- DHCP (uses UDP ports 67 and 68): udp port 67 or udp port 68
- DNS (uses UDP or TCP port 53): port 53



4. Save Captured Traffic

- When done capturing, go to File > Save As.
- Save the capture file in .pcapng or .pcap format for later analysis

5. Analyze Captured Traffic

Wireshark provides powerful tools to filter and analyze packets:

Display Filters:

You can filter displayed packets without limiting the saved capture.

Examples:

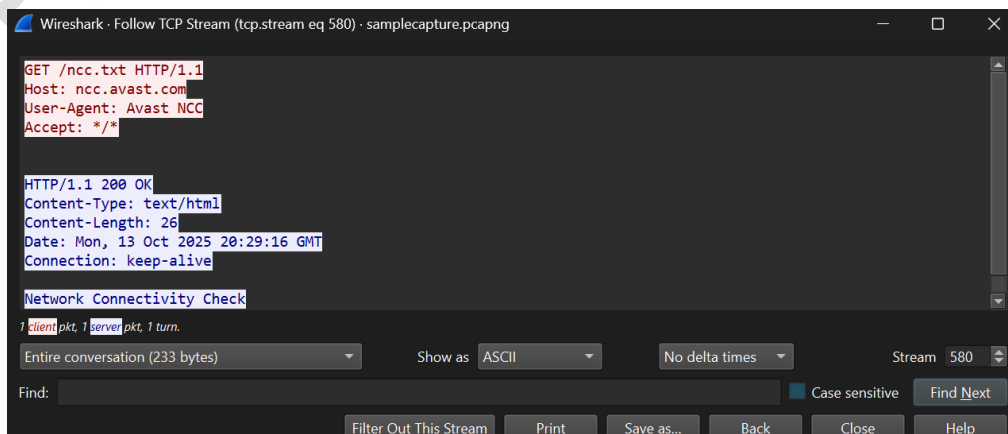
- TCP: tcp
- UDP: udp

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- The image shows a Wireshark packet capture analysis of an HTTP GET request. The top bar indicates the capture is on the 'http' interface. The packet list pane shows a list of captured packets, with packet 44399 selected. The packet details pane shows the structure of the selected packet, including Ethernet II, Internet Protocol Version 4, Transmission Control Protocol, and Hypertext Transfer Protocol. The packet bytes pane shows the raw data of the selected packet, including the Ethernet II header, IP header, TCP header, and HTTP GET request.
- | No. | Time | Source | Destination | Protocol | Length | Info |
|-------|------------|---------------------------|---------------------------|----------|--------|-----------------------------|
| 44399 | 390.711631 | 192.168.1.3 | 23.201.53.170 | HTTP | 136 | GET /ncc.txt HTTP/1.1 |
| 44399 | 390.711871 | 192.168.1.3 | 23.201.53.170 | HTTP | 136 | GET /ncc.txt HTTP/1.1 |
| 44399 | 390.718894 | 23.201.53.170 | 192.168.1.3 | HTTP | 205 | HTTP/1.1 200 OK (text/html) |
| 44400 | 390.718894 | 23.201.53.170 | 192.168.1.3 | HTTP | 205 | HTTP/1.1 200 OK (text/html) |
| 44414 | 390.727302 | 2401:4900:1cd4:3d11::2600 | 2401:4900:1cd4:3d11::2600 | HTTP | 156 | GET /ncc.txt HTTP/1.1 |
| 44415 | 390.727382 | 2401:4900:1cd4:3d11::2600 | 2401:4900:1cd4:3d11::2600 | HTTP | 156 | GET /ncc.txt HTTP/1.1 |
| 44416 | 390.727585 | 2401:4900:1cd4:3d11::2600 | 2401:4900:1cd4:3d11::2600 | HTTP | 156 | GET /ncc.txt HTTP/1.1 |
| 44417 | 390.727773 | 2401:4900:1cd4:3d11::2600 | 2401:4900:1cd4:3d11::2600 | HTTP | 156 | GET /ncc.txt HTTP/1.1 |
| 44422 | 390.736116 | 2600:140f:4200::17c | 2401:4900:1cd4:3d11::2600 | HTTP | 225 | HTTP/1.1 200 OK (text/html) |
| 44423 | 390.736116 | 2600:140f:4200::17c | 2401:4900:1cd4:3d11::2600 | HTTP | 225 | HTTP/1.1 200 OK (text/html) |
| 44424 | 390.736116 | 2600:140f:4200::17c | 2401:4900:1cd4:3d11::2600 | HTTP | 225 | HTTP/1.1 200 OK (text/html) |
| 44425 | 390.736116 | 2600:140f:4200::17c | 2401:4900:1cd4:3d11::2600 | HTTP | 225 | HTTP/1.1 200 OK (text/html) |
- Packet 44399 details:
- Ethernet II, Src: LiteonTechno_4d:fe:7f (c0:35:32:4d:fe:7f), Dst: 08:00:00:00:00:00
 - Internet Protocol Version 4, Src: 192.168.1.3, Dst: 23.201.53.170
 - Transmission Control Protocol, Src Port: 64907, Dst Port: 80, Seq: 300000000, Win: 65535, Len: 0
 - Hypertext Transfer Protocol
- Packet 44399 bytes:
- ```

3c 46 45 1f de 3a c0 35 32 4d fe 7f 08 00 45 00 <FE...5
00 7a 68 9d 40 00 80 06 82 c2 c0 a8 01 03 17 c9 zh@...
35 aa fd 8b 00 50 76 07 95 3d 44 f4 e4 38 50 18 5...Pv
00 ff ea db 00 00 47 45 54 20 2f 6e 63 63 2e 74GE
78 74 20 48 54 54 50 2f 31 2e 31 0d 0a 48 6f 73 xt HTTP/
74 3a 20 6e 63 63 2e 61 76 61 73 74 2e 63 6f 6d t: ncc.a
0d 0a 55 73 65 72 2d 41 67 65 6e 74 3a 20 41 76 ..User-A
61 73 74 20 4e 43 43 0d 0a 41 63 63 65 70 74 3a ast NCC
20 2a 2f 2a 0d 0a 0d 0a */*...

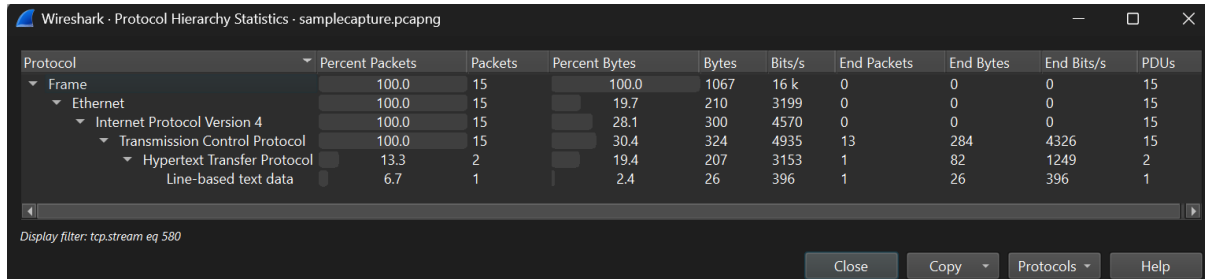
```
- Summary: Packets: 46659 - Displayed: 12 (0.0%) - Dropped: 0 (0.0%) Profile: Default

For TCP or UDP conversations, right-click a packet and select Follow > TCP Stream or Follow > UDP Stream to see the conversation in readable form.



## Protocol Hierarchy:

Use Statistics > Protocol Hierarchy to get an overview of protocols present.



Wireshark - Protocol Hierarchy Statistics - samplecapture.pcapng

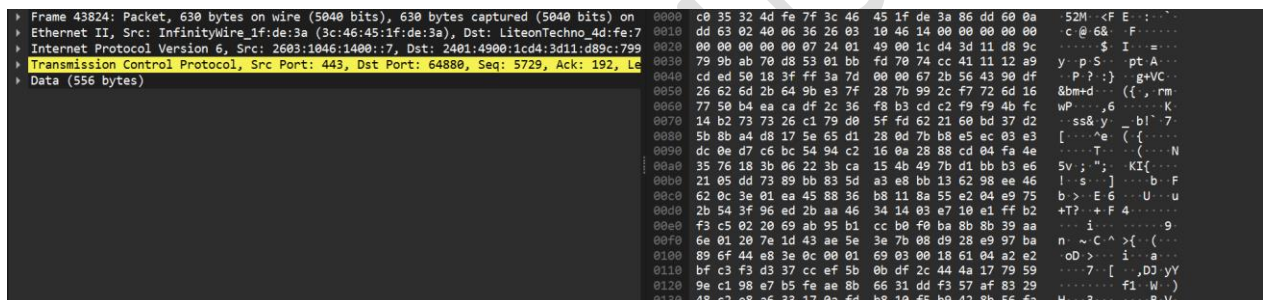
| Protocol                      | Percent Packets | Packets | Percent Bytes | Bytes | Bits/s | End Packets | End Bytes | End Bits/s | PDUs |
|-------------------------------|-----------------|---------|---------------|-------|--------|-------------|-----------|------------|------|
| Frame                         | 100.0           | 15      | 100.0         | 1067  | 16 k   | 0           | 0         | 0          | 15   |
| Ethernet                      | 100.0           | 15      | 19.7          | 210   | 3199   | 0           | 0         | 0          | 15   |
| Internet Protocol Version 4   | 100.0           | 15      | 28.1          | 300   | 4570   | 0           | 0         | 0          | 15   |
| Transmission Control Protocol | 100.0           | 15      | 30.4          | 324   | 4935   | 13          | 284       | 4326       | 15   |
| Hypertext Transfer Protocol   | 13.3            | 2       | 19.4          | 207   | 3153   | 1           | 82        | 1249       | 2    |
| Line-based text data          | 6.7             | 1       | 2.4           | 26    | 396    | 1           | 26        | 396        | 1    |

Display filter: tcp.stream eq 590

Buttons: Close, Copy, Protocols, Help

## Packet Details:

Select any packet and expand protocol layers in the middle pane to analyze headers and payload details.



Wireshark - Packet Details - samplecapture.pcapng

Frame 43824: Packet, 630 bytes on wire (5040 bits), 630 bytes captured (5040 bits) on interface 0

Ethernet II, Src: InfinityWire\_1f:de:3a (3c:46:45:1f:de:3a), Dst: LiteonTechno\_4d:fe:7f (08:00:27:4d:fe:7f)

Internet Protocol Version 6, Src: 2603:1046:1400::7, Dst: 2401:4900:1cd4:3d11:d89c:799

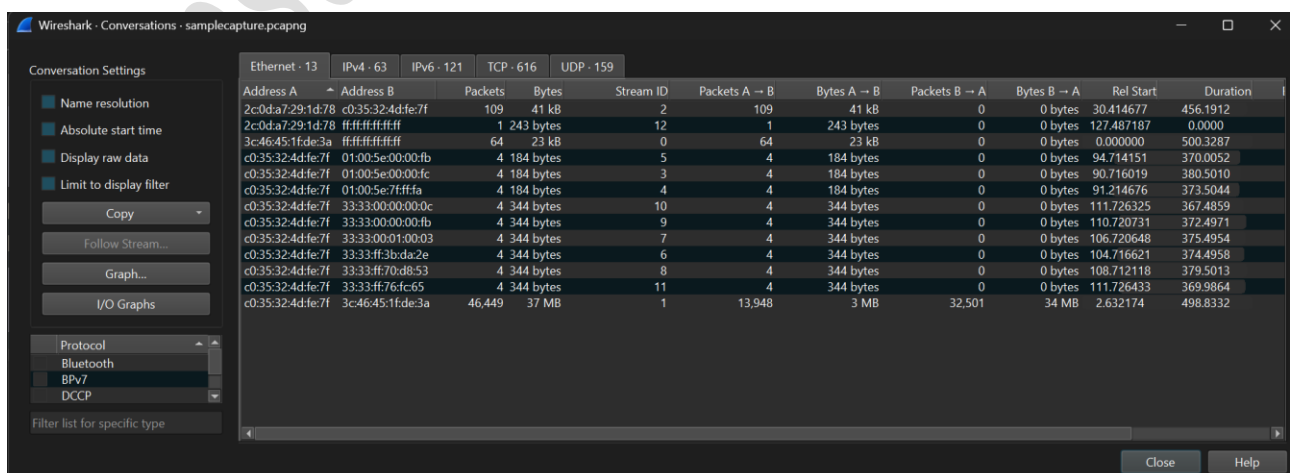
Transmission Control Protocol, Src Port: 443, Dst Port: 64880, Seq: 5729, Ack: 192, Len: 556

Data (556 bytes)

Hex and ASCII view of the packet payload.

## Conversations:

Use Statistics > Conversations to analyze communication between source and destination hosts.



Wireshark - Conversations - samplecapture.pcapng

Conversation Settings

- Name resolution
- Absolute start time
- Display raw data
- Limit to display filter
- Copy
- Follow Stream...
- Graph...
- I/O Graphs

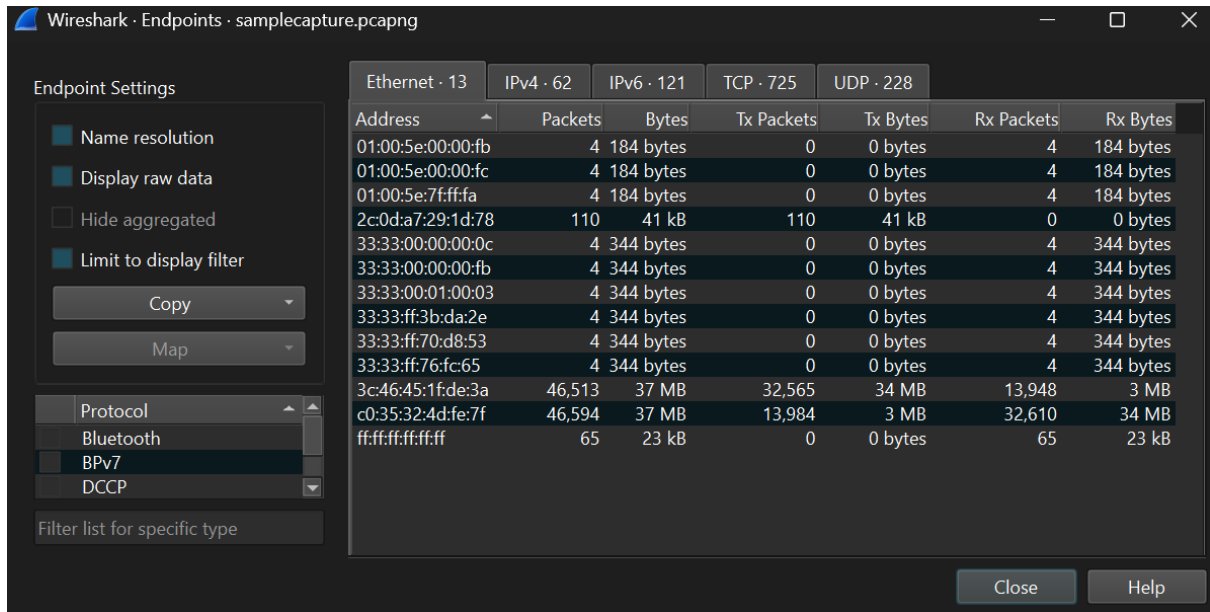
Protocol: Bluetooth, IPv6, DCCP

| Address A         | Address B         | Packets | Bytes     | Stream ID | Packets A → B | Bytes A → B | Packets B → A | Bytes B → A | Rel Start  | Duration |
|-------------------|-------------------|---------|-----------|-----------|---------------|-------------|---------------|-------------|------------|----------|
| 2c:0d:a7:29:1d:78 | c0:35:32:4d:fe:7f | 109     | 41 kB     | 2         | 109           | 41 kB       | 0             | 0 bytes     | 30.414677  | 456.1912 |
| 2c:0d:a7:29:1d:78 | ff:ff:ff:ff:ff:ff | 1       | 243 bytes | 12        | 1             | 243 bytes   | 0             | 0 bytes     | 127.487187 | 0.0000   |
| 3c:46:45:1f:de:3a | ff:ff:ff:ff:ff:ff | 64      | 23 kB     | 0         | 64            | 23 kB       | 0             | 0 bytes     | 0.000000   | 500.3287 |
| c0:35:32:4d:fe:7f | 01:00:5e:00:00:fb | 4       | 184 bytes | 5         | 4             | 184 bytes   | 0             | 0 bytes     | 94.714151  | 370.0052 |
| c0:35:32:4d:fe:7f | 01:00:5e:00:00:fc | 4       | 184 bytes | 3         | 4             | 184 bytes   | 0             | 0 bytes     | 90.716019  | 380.5010 |
| c0:35:32:4d:fe:7f | 01:00:5e:7f:ff:fa | 4       | 184 bytes | 4         | 4             | 184 bytes   | 0             | 0 bytes     | 91.214676  | 373.5044 |
| c0:35:32:4d:fe:7f | 33:33:00:00:00:0c | 4       | 344 bytes | 10        | 4             | 344 bytes   | 0             | 0 bytes     | 111.726325 | 367.4859 |
| c0:35:32:4d:fe:7f | 33:33:00:00:00:fb | 4       | 344 bytes | 9         | 4             | 344 bytes   | 0             | 0 bytes     | 110.720731 | 372.4971 |
| c0:35:32:4d:fe:7f | 33:33:00:01:00:03 | 4       | 344 bytes | 7         | 4             | 344 bytes   | 0             | 0 bytes     | 106.720648 | 375.4954 |
| c0:35:32:4d:fe:7f | 33:33:ff:3b:da:2e | 4       | 344 bytes | 6         | 4             | 344 bytes   | 0             | 0 bytes     | 104.716621 | 374.4958 |
| c0:35:32:4d:fe:7f | 33:33:ff:70:d8:53 | 4       | 344 bytes | 8         | 4             | 344 bytes   | 0             | 0 bytes     | 108.712118 | 379.5013 |
| c0:35:32:4d:fe:7f | 33:33:ff:76:fc:65 | 4       | 344 bytes | 11        | 4             | 344 bytes   | 0             | 0 bytes     | 111.726433 | 369.9864 |
| c0:35:32:4d:fe:7f | 3c:46:45:1f:de:3a | 46,449  | 37 MB     | 1         | 13,948        | 3 MB        | 32,501        | 34 MB       | 2.632174   | 498.8332 |

Buttons: Close, Help

## Endpoints:

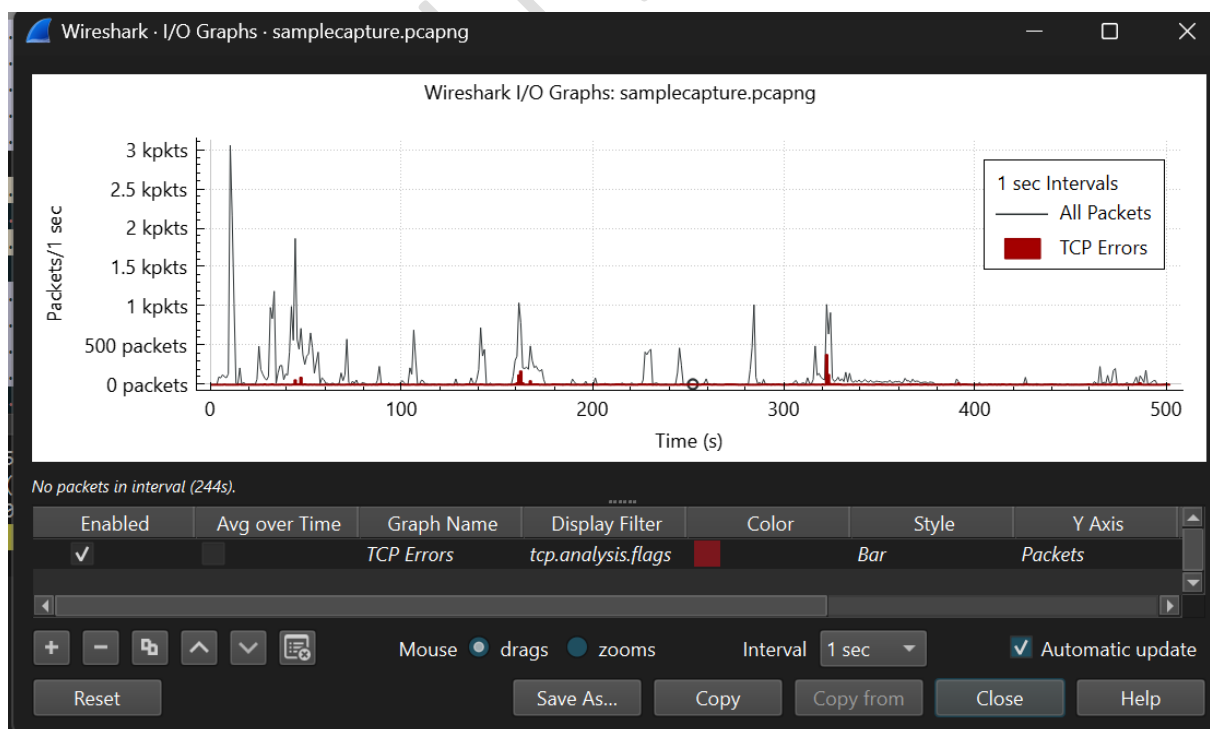
Go to Statistics > Endpoints to list all active IP/MAC addresses involved in the capture.



| Address           | Packets | Bytes     | Tx Packets | Tx Bytes | Rx Packets | Rx Bytes  |
|-------------------|---------|-----------|------------|----------|------------|-----------|
| 01:00:5e:00:00:fb | 4       | 184 bytes | 0          | 0 bytes  | 4          | 184 bytes |
| 01:00:5e:00:00:fc | 4       | 184 bytes | 0          | 0 bytes  | 4          | 184 bytes |
| 01:00:5e:7f:ff:fa | 4       | 184 bytes | 0          | 0 bytes  | 4          | 184 bytes |
| 2c:0d:a7:29:1d:78 | 110     | 41 kB     | 110        | 41 kB    | 0          | 0 bytes   |
| 33:33:00:00:00:0c | 4       | 344 bytes | 0          | 0 bytes  | 4          | 344 bytes |
| 33:33:00:00:00:fb | 4       | 344 bytes | 0          | 0 bytes  | 4          | 344 bytes |
| 33:33:00:01:00:03 | 4       | 344 bytes | 0          | 0 bytes  | 4          | 344 bytes |
| 33:33:ff:3b:da:2e | 4       | 344 bytes | 0          | 0 bytes  | 4          | 344 bytes |
| 33:33:ff:70:d8:53 | 4       | 344 bytes | 0          | 0 bytes  | 4          | 344 bytes |
| 33:33:ff:76:fc:65 | 4       | 344 bytes | 0          | 0 bytes  | 4          | 344 bytes |
| 3c:46:45:1f:de:3a | 46,513  | 37 MB     | 32,565     | 34 MB    | 13,948     | 3 MB      |
| c0:35:32:4d:fe:7f | 46,594  | 37 MB     | 13,984     | 3 MB     | 32,610     | 34 MB     |
| ff:ff:ff:ff:ff:ff | 65      | 23 kB     | 0          | 0 bytes  | 65         | 23 kB     |

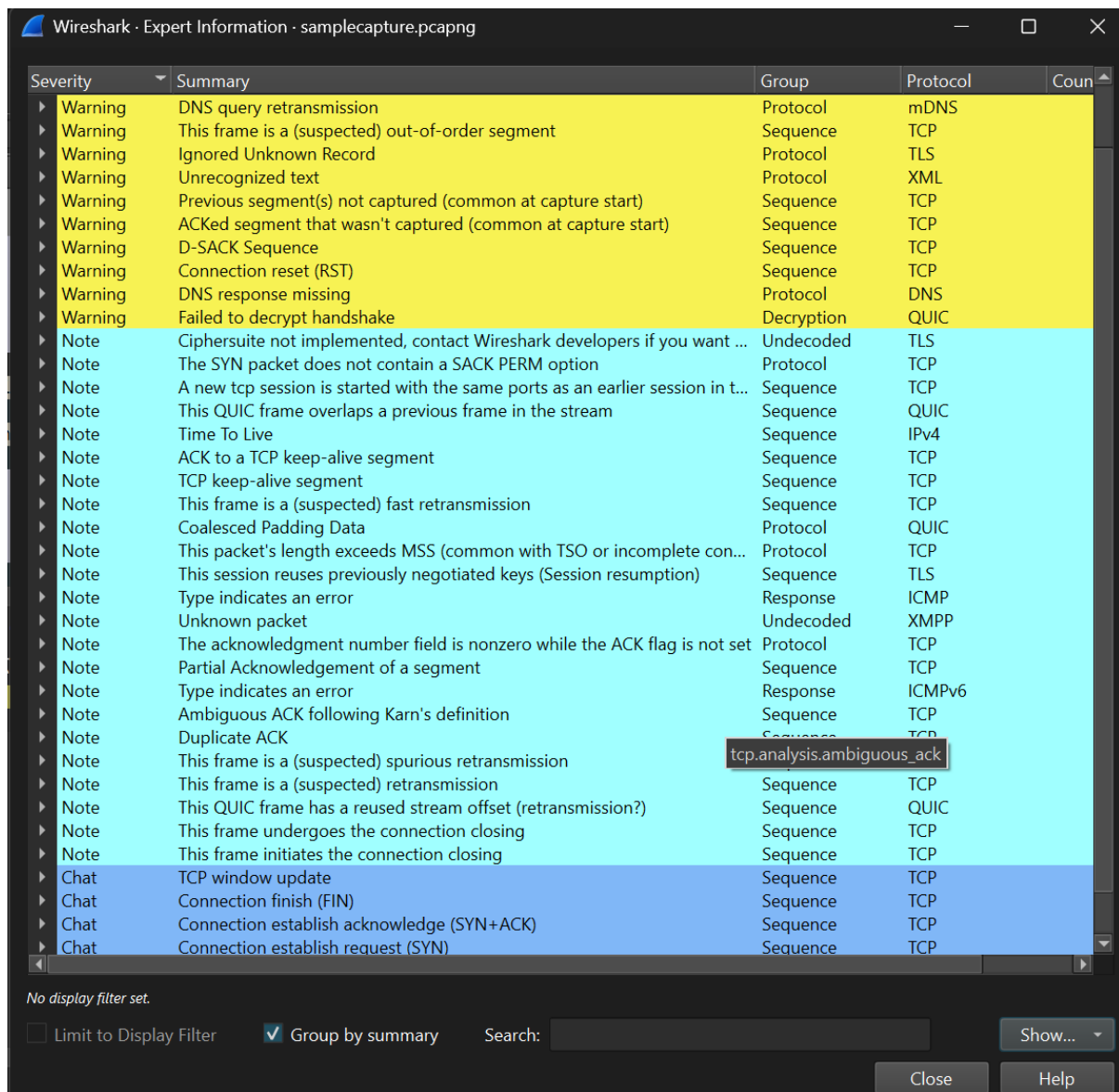
## I/O Graphs:

Use Statistics > I/O Graphs to visualize traffic rate and identify spikes or drops over time.



## Expert Information:

Open Analyze > Expert Information to detect errors, warnings, and retransmissions.



## Result:

Thus network traffic on TCP/UDP/IP/HTTP/ARP/DHCP/ICMP /DNS were analyzed using Wireshark Tool.